

ARSITHA SATHU

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SUMMARY

Software Engineer with a Master's in Computer Science and 2+ years of experience building scalable production applications. Specialized in AI Engineering, architecting Agentic Workflows (MCP), RAG engines, and low-latency voice agents using Python and Azure AI. Expert in full-stack development with Java (Spring Boot), React, and PostgreSQL, with a focus on API design and writing maintainable, production-ready code.

EDUCATION

MASTER'S DEGREE | Computer Science | George Mason University | GPA – 3.83

Aug 2023 – May 2025

BACHELOR'S DEGREE | Computer Science Engineering | Jawaharlal Technological University | CGPA – 9.21

Aug 2017 – June 2021

TECHNICAL SKILLS

Languages: Java, Python, JavaScript, TypeScript, SQL, C, HTML, CSS, Bash, shell scripting

Frameworks & Libraries: Spring Boot, React.js, Node.js, Angular, FastAPI, PyTorch, OpenAI SDK, LangChain (Agentic Workflows), PyPDF

Data & Infrastructure: PostgreSQL, MySQL, MongoDB, REST API, Microservices, WebSockets, AWS (EC2, S3), Azure (AI Foundry, Speech Services, Container Apps, OpenAI (GPT-4o), Azure Key Vault), Docker, Kubernetes, Jenkins, CI/CD, RAG, MCP, AudioWorklet

Development Tools: Git, Postman, Swagger, JUnit, JIRA, SonarQube, VS Code, IntelliJ IDEA, ChatGPT, Claude, GitHub Copilot, Cursor, Gemini

Domain Knowledge: Data Structures & Algorithms, Operating Systems, Embedded Systems, NLP, LLMs, Data Mining, Networking Fundamentals, TCP/IP Model, UDP, HTTP/HTTPS

PROFESSIONAL EXPERIENCE

BLUE HORSE DIGITAL | Software Engineer | Remote

July 2025 – present

- Architected a low-latency, voice agent using Python (aiohttp) and Azure AI Services (Speech/GPT-4o), optimizing client-side audio with a custom JavaScript Audio Worklet (resampling/anti-aliasing) and implementing barge-in interruption for sub-second, natural responsiveness.
- Built a custom Agentic Workflow (MCP-based) and Retrieval-Augmented Generation (RAG) engine leveraging Azure OpenAI Embeddings to autonomously execute complex reasoning, query USDA policies, and provide accurate, citation-backed answers.
- Engineered a hands-free PDF Form Filling system that dynamically inspects schemas (using pypdf) to map conversational data into applications in real-time, auto-generating and emailing completed forms without manual input.
- Deployed the scalable, containerized application to Azure Container Apps using Docker multi-stage builds, ensuring secure API key management and robust production infrastructure.

INFOSYS LIMITED | Full-stack Developer | India

Oct 2021 – July 2023

- Developed production REST APIs and microservices using Java, Spring Boot, Angular, and PostgreSQL, supporting enterprise applications across multiple client projects.
- Optimized database queries and API endpoints, improving response times by 25-30% through strategic indexing and query refactoring.
- Built a reusable Angular component library (dynamic tables, filters, role-based controls) adopted across development teams, reducing UI development time by 30%.
- Resolved high-priority production bottlenecks with over 90% test coverage using JUnit while driving continuous integration and code quality within an Agile Scrum methodology.

PROJECTS

AI LOG ANALYSIS | Python, React, JavaScript, Spring Boot, GPT-4o-mini, PostgreSQL, Git

- Architected a microservice-based log analysis platform with a Spring Boot backend, a Python FastAPI AI service, and a React frontend for automated error detection and debugging.
- Built a log ingestion pipeline that parses application logs, stores them in PostgreSQL via Spring Data JPA, and triggers AI analysis through WebClient REST integration.
- Developed an interactive React dashboard for one-click log analysis, visualizing GPT-4o-mini generated error clusters, root cause explanations, and actionable fix recommendations.

ABSTRACTIVE TEXT SUMMARIZATION | Python, mT5 Transformer, Git, PyTorch, Transformers, Wandb, Pandas, NumPy

- Fine-tuned an mT5 Transformer on a custom dataset of 200+ Telugu narratives, implementing a specialized NLP preprocessing pipeline and achieving optimized performance tracked via ROUGE metrics and Wandb.

DEPARTMENTAL CONNECT | ReactJS, HTML, CSS, Bootstrap, JavaScript, Node.js, SQL, Git

- Developed a dynamic full-stack application using React, Node.js, and MySQL, resulting in a 70% improvement in load times and a 45% increase in user engagement.

ACHIEVEMENTS AND ACTIVITIES

- Received Outstanding Academic Achievement Award during my master's in computer science at George Mason University.
- Won 1st place in the CSE Department Webathon technical competition during my bachelor's.
- Held responsibilities as a Department Magazine Co-Ordinator and class representative during my undergraduate studies for 4 years.